DRUGS · BLEEDING AND HAEMOSTASIS

Reversal of anticoagulation

Reversal agents restore clotting when an anticoagulated patient has life-threatening bleeding or needs emergency surgery: a massive epistaxis, a secondary tonsillar haemorrhage, an expanding neck haematoma, a bleeding tumour. Each anticoagulant has its own reversal strategy, and haematology advice should be sought early.

1 CHAPTER ONE At a glance

The agents on one page.

1.1 Summary

REVERSAL OF ANTICOAGULATION

	SUMMARY
Warfarin	Life-threatening bleeding: four-factor prothrombin complex concentrate (PCC) 25 to 50 units/kg (by INR and weight) plus vitamin K 5 to 10 mg intravenously; non-urgent high INR without bleeding: oral vitamin K 1 to 5 mg and withhold doses
Dabigatran	Idarucizumab 5 g intravenously (two 2.5 g doses); haemodialysis removes it
Apixaban, rivaroxaban, edoxaban	Andexanet alfa where available and indicated; otherwise PCC 25 to 50 units/kg; activated charcoal if taken within a few hours
Unfractionated heparin	Protamine 1 mg per 100 units given in the previous 2 to 3 hours (maximum 50 mg)
Low molecular weight heparin	Protamine gives partial reversal (about 60% of anti-Xa activity)
Antiplatelets	No specific reversal; platelet transfusion in life-threatening bleeding or emergency surgery; desmopressin in some cases

	SUMMARY
Adjuncts	Tranexamic acid; local measures; treat hypothermia, acidosis and hypocalcaemia in major haemorrhage

2 CHAPTER TWO Using them in ENT

Practice.

2.1 The bleeding patient on an anticoagulant

Resuscitate, control the bleeding locally (packing, pressure, theatre), take blood for full blood count, clotting, renal function and group and save or crossmatch, establish which drug was taken and when, and involve haematology. Restart anticoagulation once bleeding is controlled and the risk allows, usually within days, because the thrombotic risk persists.

3 CHAPTER THREE Safety

What goes wrong.

3.1 Adverse effects

- PCC and andexanet: **thrombosis**.
- Vitamin K intravenously: rare anaphylactoid reactions; give slowly. High doses make re-warfarinisation difficult.
- Protamine: hypotension, bradycardia, anaphylaxis (fish allergy, previous protamine-containing insulin, vasectomy).

4 CHAPTER FOUR For the examination

What the examiner wants to hear.

4.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Warfarin reversal**: PCC plus intravenous vitamin K.
- Specific reversal for **dabigatran** (idarucizumab) and the factor Xa inhibitors.
- Protamine dosing.

4.2 Common pitfalls

COMMON PITFALLS

- Using fresh frozen plasma alone for warfarin reversal in life-threatening bleeding.
- Giving vitamin K without PCC in an emergency (it takes hours to work).

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